

Network Equipment Communication Protocol

Message Specification V1.0

Revision History

Version	Creation/Revision Date	Author/Modifier	Revision Description
1.0	2017-03-28	dylan	Added
1.1	2017-04-07	dylan	Corrected some errors
1.2	2017-04-10	Dylan	Added the TFLV structure description
1.3	2017-06-29	Dylan	Added some parameters
1.4	2017-11-20	Dylan	Added IMEI number support and the Do Not Disturb function
1.5	2018-01-05	Dylan	Added registration parameter - Parameter Synchronization Identifier Added control parameter - Restore Factory Settings
1.6	2018-05-03	Liya	Added configuration parameter - Early Medication Retrieval Validity Period Added configuration parameter - Alarm repeat function setting Added control parameter - Early Medication Retrieval
1.7	2018-09-20	Liya	Added Event Parameters-Pill Tray Reset Report
1.8	2018-11-13	Liya	Added IoT SIM card CCID reporting Added network configuration result reporting
1.9	2018-12-12	dylan	Added User Binding Relationship(0x800A) Added Factory Reset Report(0xC501)
1.10	2018-12-18	dylan	Added Device Startup Report(0xC206) Added Device Startup Report(0xC601)
1.11	2019-01-14	Dylan	Added Medication Retrieval Trigger Identifier(0xC207)
1.12	2019-01-17	Dylan	Added Medication Retrieval Time(0xC208)
1.13	2019-03-22	dylan	Added Online Hold Time(0x800B) Added Set Heartbeat Interval(0xA011) Added Set Online Hold Time(0xA012)

1.14	2019-06-20	Dylan	The definitions of “Byte Array” and “String” were exchanged; existing parameters similar to byte arrays were changed to the String type.
1.15	2019-12-06	Dylan	Protocol optimization
1.16	2020-01-15	Dylan	Alarm duration (0x1011) unit changed from seconds to minutes; Added parameter - Additional Function Identifier (0x8081) Added parameter - Custom Number (0x8082) Removed the reserved parameter - Set Online Hold Time (0xA012) Added parameter - Set Server IP (0xA021) Added parameter - Set Server Domain Name (0xA022) Added parameter - Set Server Port (0xA023) The value definition for Early Medication Retrieval (0xA123) was removed Added parameter - Device Re-registration (0xA004)
1.17	2020-06-16	Dylan	Added alarm parameters: 0x1027, 0x1028, 0x1029, 0x1037, 0x1038, and 0x1039
1.18	2022-05-13	Dylan	Added parameter - Medication Status (0x8101) Added parameter - DC Power Status (0x8109) Added parameter - Pill Tray Reset Fault (0x8122) Added parameter - Emergency Call Status (0x8123) Added parameters - Alarm Status 1-9 (0x8131-0x8139) Added parameter - Emergency Call Trigger Time (0x1061) Added parameter - Key Tone Switch (0x1062) Added parameter - Alarm Ringtone Type (0x1012) Removed parameter - Alarm Ringtone Hold Duration (0x1011)
1.19	2022-07-25	Dylan	Added parameter value - Battery Status (0x8104): Value 4 - No battery
1.20	2022-08-29	Dylan	Reorganized parameter tables
1.21	2022-09-05	Dylan	Added all M2 parameters (device type 0x02)

1.22	2022-09-08	Dylan	Added upgrade-related packet types (0x0E, 0x8E, 0x0F, 0x8F)
1.23	2022-09-27	Dylan	Added a parameter for the remote upgrade function (upgrade timeout)
1.24	2022-10-28	Dylan	Changed the M2 Battery Status values (0x8104)
1.25	2024-05-06	ZhouZS	Corrected an error in the horizontal message-structure illustration
1.26	2025-12-19	ZhouZS	Added parameter - Alarm Ring Pause Duration (0x1063) Added parameter - Rotate to Specified Medication Compartment (0xA124) Added parameter - Medication Pause Duration Setting (0xA125) Added parameter - Pill Tray Rotation Status (0x8140) Added parameter - Medication Pause Status (0x8141) Adjusted parameter for M1/M2 consistency - Alarm Ringtone Type (0x1012) Adjusted parameter for M1/M2 consistency - Volume Level (0x1013) Adjusted parameter for M1/M2 consistency - Medication Retrieval Warning Time (0x1017) Adjusted parameter for M1/M2 consistency - Missed Medication Retrieval Time (0x1018) Adjusted parameter for M1/M2 consistency - Current Medication Compartment Number (0x101A) Adjusted parameter for M1/M2 consistency - Battery Level (0x8103) Adjusted parameter for M1/M2 consistency - Battery Status (0x8104)

Terminology

Proper Noun		Definition
Byte Sequence		Refers to the storage order of bytes when multi-byte data is stored in computer memory or transmitted over the network.
Host Byte Order		The low address stores the low byte of word data, and the high address stores the high byte of word data.
Network Order	Byte	The high byte of word data is stored in the low address, while the low byte of word data is stored in the high address

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1. Message Specification

1.1. Data Type Definition

1.1.1. Data Type Definition

Data Type	Definition
INT8U	Unsigned integer (1 byte)
INT8S	Signed 8-bit integer (1 byte)
INT16U	Unsigned integer (2 bytes)
INT16S	Signed integer(2 bytes)
INT32U	Unsigned integer(4 bytes)
INT32S	Signed integer(4 bytes)
INT64U	Unsigned integer(8 bytes)
INT64S	Signed integer(8 bytes)
FLOAT	Single-precision floating-point number
DOUBLE	Double-precision floating-point number
STRING	Character string
ARRAY	Byte array

1.1.2. Data Processing Rules

- The content to be filled in for each field is described in the sections for the corresponding packet types.
- When “Length” is n, the length of this field is fixed at n bytes.
When “Length” is [m,n], the minimum length of this field is m bytes and the maximum length is n bytes.
- When “Mandatory/Optional” is M, this element is mandatory and must be present.
When “Mandatory/Optional” is O, this element is optional and should be filled in as required.

1.1.3. Byte Order

The byte order for data transmission in this protocol is host byte order.

2. Message Format Description

2.1. Message Structure Description

The basic message structure is shown in the table below:

Header	Length	Status	Version	Serial Num ber	Subserial number	Subpacket amount	Flag	Device Type	Device number	Packet Type	Application Data	Check code
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The structure fields are described in the table below:

Name	Instruction	Data Type	Length	Mandatory/ Optional
Header	Fixed to 0xAA	INT8U	1	M
Length	Length of total bytes amount (From “Status” to “Application Data”)	INT16U	2	M
Status	Status of the data packet, value refers to "Status Identification"	INT8U	1	M
Version	Protocol version, fixed to 0x01	INT8U	1	M
Serial number	Packet sequence number, incremented each time a new packet is sent, 0~65535	INT16U	2	M
Subserial number	A large data packet is split into several sub-packets and sent. Identifies the sequence number of the subpacket. The sequence number starts from 0.	INT8U	1	M
Subpacket total amount	A large data packet is split into several sub-packets and sent. Identifies the total number of subpackets	INT8U	1	M
Flag	Various functional parameters that identify the data packet. For values, see "Functional Identification"	INT8U	1	M
Device type	Device type, see value Device type	INT8U	1	M

Device number	Unique identifiers for different devices. For values, see "device number"	INT64U	8	M
Packet type	Packet type, please refer to "《Packet type description》" for the value.	INT8U	1	M
Application Data	Indicates the application data, see the description of each packet type for the specific format.	INT8U	[0,1400]	O
Check code	Indicates that the data from the "Length" to the end of the "Application data" is data-checked, and the checking algorithm is CRC16; for the implementation of Algorithm C, see "CRC16 Algorithm".	INT16U	2	M

2.2. TFLV Structure Description

Purpose: used to convey the value of an element in the protocol.

The basic structure is shown in the table below:

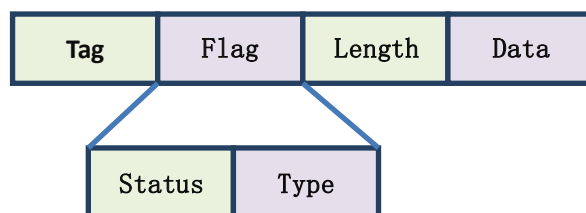


Figure 2.2-12.2

The structure fields are described in the table below:

Field Name	Field Description	Field Type (see 1.1.1 Data Type Definition")	Length	Mandatory/Optional
Tag	The number of the information carried by this TFLV.	INT16U	2	M
Flag	Bits 0~4 indicate the data type. For values, see "Appendix - Code List - TFLV Structure - Parameter Type". Bits 5~7 represent status, see the value "Appendix - Code List - TFLV Structure - Status Identification"	INT8U	1	M
Length	Length of data value	INT8U	1	M

Data Value	Data information	ARRAY	[0,255]	O
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3. Message Transmission Protocol

TCP, UDP, and HTTP are supported, with TCP preferred.

When using HTTP, the content-type is application/octet-stream.

4. Message Application Processing Modes

4.1. Mode 1: Synchronous No-Response Processing Mode



Figure 4.1-14.11

This mode is used when Terminal A actively sends data to the device and does not require a response from the device.

4.2. Mode 2: Synchronous Response Processing Mode

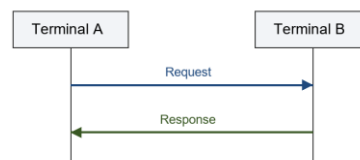


Figure 4.2-24.22

This mode is used when Terminal A sends data to the device and requires Terminal B to respond. If Terminal B does not respond within the specified time, Terminal A retransmits the packet. If the device still does not respond after the specified number of retransmissions, communication between Terminals A and B is considered abnormal and the connection should be disconnected.

In this mode, the data protocol requires each packet to stay within the maximum data length specified by the application protocol. When the total length of a request or response exceeds that maximum, the data must be split into subpackets. Therefore, three cases must be considered:

4.2.1. One request requires one request packet, and one request packet requires one response packet

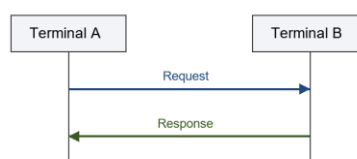


Figure 4.2-24.2

After one interaction is completed, refer to the description of “Mode 2: Synchronous Response Processing Mode” .

In this mode, the subpacket information fields (subpacket sequence number and total number of subpackets) are both set to 0.

4.2.2. One request requires one request packet, and one request packet requires n response packets

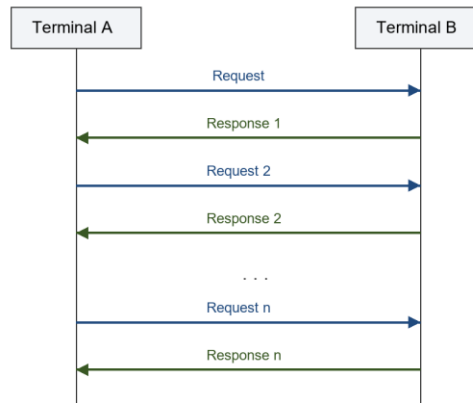


Figure 4.2-44.2

Terminal A sends a request and receives a response from Terminal B. If parsing the response shows that the response data requires multiple response packets, the subpacket fields in the previous request packet must be updated to identify the response sequence requested, and the packet must be sent to Terminal B again.

1. Requester

When sending a request packet, set both subpacket information fields (subpacket sequence number and total number of subpackets) to 0.

When a response packet is received and the total number of subpackets is not 0, the response requires n response packets. Increment the subpacket sequence number in the original request packet by 1, set the request packet total number of subpackets to the value in the response packet, and continue sending request packets. When the subpacket sequence number plus 1 equals the total number of subpackets, the response has been fully received.

2. Sender

When a request packet is received with both the total number of subpackets and the subpacket sequence number set to 0, determine whether the response requires multiple response packets. If so, send the first packet and set the subpacket data in the request packet to the total number of response packets, informing the requester of the total number of subpackets to be returned.

When the total number of subpackets is not 0, return the response subpacket corresponding to the subpacket sequence number in the request packet.

4.2.3. One request requires n request packets, and one request packet requires one response packet

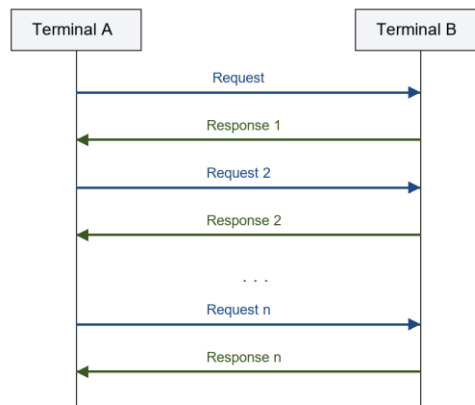


Figure 4.2-44.2

Because one request must be divided into multiple request packets, each request packet is processed as an independent request and the device must return the corresponding response information.

See “Mode 2: Synchronous Response Processing Mode - One request requires one request packet, and one request packet requires one response packet” .

5. Exception Handling

5.1. Response Timeout Exception

When the sending terminal does not receive a response from the receiving terminal within the specified time, it retransmits the data. If no response is received after the specified number of retransmissions, communication between the two terminals is considered abnormal and the connection should be disconnected.

5.2. Abnormal Packet Status Identification

If the sending terminal receives a non-zero status identifier, the previously sent packet could not be recognized by the receiving terminal. A notification should be provided.

If the message header received by the receiving terminal is abnormal, set the status identifier of the original received packet to the corresponding error identifier and return it to the sending terminal.

5.3. Unsupported Protocol Version

If a terminal receives a packet with an unsupported protocol version, it must set the packet status bit to the corresponding status and return the adjusted original packet to the sender.

5.4. Unsupported Device Type

If a request packet contains an unsupported device type, set the packet status bit to the corresponding status and return the adjusted original packet to the sender.

5.5. Unsupported Packet Type

If a request packet contains an unsupported packet type, set the packet status bit to the corresponding status and return the adjusted original packet to the sender.

5.6. Inconsistent Device ID

When a terminal receives a packet, it checks whether the device IDs match. If they do not match, set the packet status bit to the corresponding status and return the adjusted original packet to the sender.

5.7. CRC Check Exception

If a CRC error occurs in a received packet, discard the packet.

6. Packet Type Description

Packet Type	Response packet type	Sender	Related instructions
0x01	0x81	Client	Registration packet
0x02	0x82	Client	Heartbeat packet
0x03	0x83	Client	event packet
0x04	0x84	Client	Notification packet
0x05	0x85	Server	Read configuration parameters
0x06	0x86	Server	Set configuration parameters
0x07	0x87	Server	Status query
0x08	0x88	Server	Device control
0x0A	0x8A	Server	Read configuration parameter information (configuration parameter list supported by the device)
0x0B	0x8B	Server	Read status parameter information (list of status parameters supported by the device)
0x0C	0x8C	Server	Read control parameter information (list of control parameters supported by the device)
0x0D	0x8D	Server	Read event parameter information (event parameter list supported by the device)
0x0E	0x8D	Server	Upgrade start
0x0F	0x8F	Server	Upgrade data transfer

7. Interaction Modes

7.1. Interaction Description

7.1.1. Registration Phase

Client: After establishing a connection, actively send a registration packet to the Server.

If a registration response packet is required, the Client waits for it to return. When the registration response packet is received, the Client is registered successfully. If it is not received within the timeout period, the Client resends the registration packet. If the retransmission count is exceeded, registration is considered unsuccessful and the connection is disconnected and re-established.

If no registration response packet is required, registration is considered successful.

Server: after establishing a connection, wait to receive the Client registration packet.

If the Client registration packet is not received within the specified time, registration is considered unsuccessful and the connection is disconnected.

If a Client registration packet is received, parse it and check whether a link for the specified “Device ID” exists. If it does, disconnect the previous link and use the new link for communication.

When a registration response packet is required, send a registration response packet to the Client and consider the Client successfully registered.

When no registration response packet is required, consider the Client successfully registered.

If this is the Client’s first registration, the Server should query the Client’s parameter information for synchronization.

1. Registration Packet(0x01)

The data body format of the registration packet is as follows:



The TAG values recommended for TFLV are as follows:

TAG	Description	Field Type	Length	Mandatory/Optional
0x8002	Version	INT16U	2	O

0x8003	Packet Size	INT16U	2	O
0x8004	heartbeat interval	INT16U	2	O
0x8005	data encryption	INT8U	1	O
0x8006	Response timeout	INT16U	2	O
0x8007	Number of retransmissions after timeout	INT8U	1	O
0x8008	Parameter synchronization identifier	INT8U	1	O
0x8009	SIM card CCID	STRING	20	O
0x800A	User binding relationship	INT8U	1	O
0x800B	Online hold time	INT16U	2	O
0x8081	additional function identifier	INT32U	4	O
0x8082	Custom number	INT32U	4	O

2. Registration Response Packet(0x81)

The data body format of the registration response packet is as follows:

None

If registration fails, set the corresponding status in the status identifier and return the packet to the requester.

7.1.2. Running Phase

After the Client and Server complete registration, they can exchange data, control information, and other information.

3. Heartbeat Packet(0x02)

When no data is exchanged between the Server and Client for a certain period, the Client should actively send a heartbeat packet to the Server to maintain the link between them.

Client:

If a heartbeat response packet is required, wait for it to return. Receipt of the response

indicates successful transmission. If no response is received within the timeout period, resend the packet. If the retransmission count is exceeded, transmission is considered unsuccessful and the connection is disconnected and re-established.

If no heartbeat response packet is required, completion of transmission indicates successful transmission.

The data body format of the heartbeat packet is as follows:



TFLV can carry status parameters and report status at the same time.

4. Heartbeat Response Packet(0x82)

When the Server receives a heartbeat packet, the link with the Client is considered normal. If a response is required, send a heartbeat response packet.

The heartbeat response packet format is as follows:

None

5. Event Packet(0x03)

When the Client needs to inform the Server of specific information, the Client should actively send an event packet to the Server.

Client:

If a response packet is required, wait for it to return. Receipt of the response indicates successful transmission. If no response is received within the timeout period, resend the packet. If the retransmission count is exceeded, transmission is considered unsuccessful and the connection is disconnected and re-established.

If no response packet is required, completion of transmission indicates successful transmission.

The data body format of the event packet is as follows:



The TAG values that TFLV can carry are based on the “TAG Definition for Each Device Type” ..

6. Event Response Packet(0x83)

When the Server receives an event packet, it processes the related information. If a response is required, it sends a response packet.

The event response packet format is as follows:



The TAG values that TFLV can carry are based on the “TAG Definition for Each Device Type” ..

7. Notification Packet(0x04)

When the Client needs to notify the Server of changes to configuration or status parameters, the Client should actively send a notification packet to the Server.

Client:

If a response packet is required, wait for it to return. Receipt of the response indicates successful transmission. If no response is received within the timeout period, resend the packet. If the retransmission count is exceeded, transmission is considered unsuccessful and the connection is disconnected and re-established.

If no response packet is required, completion of transmission indicates successful transmission.

The data body format of the notification packet is as follows:



The TAG values that TFLV can carry are based on the “TAG Definition for Each Device Type” ..

8. Notification Response Packet(0x84)

When the Server receives a notification packet, it processes the related information. If a response is required, it sends a response packet.

The notification response packet format is as follows:



The TAG values that TFLV can carry are based on the “TAG Definition for Each Device Type” ..

9. Read Configuration Parameter Request Packet(0x05)

When the Server needs to read configuration parameters from the Client, the Server should actively send a Read Configuration Parameter Request Packet to the Client.

In the TFLV structure, set Length and Type according to the corresponding values for the Tag, and fill Data Value with 0x00 for the corresponding length.

Server:

The read configuration parameter request packet requires waiting for a response. Receipt of the response indicates successful transmission. If no response is received within the timeout period, resend the request packet. If the retransmission count is exceeded, transmission is considered unsuccessful and the connection is disconnected and re-established.

The format of the Configuration parameter request packet is as follows:



The TAG values that TFLV can carry are based on the “TAG Definition for Each Device Type” ..

10. Read Configuration Parameter Request Response

Packet(0x85)

When the Client receives a Read Parameter Request Packet, it parses the packet and checks the CRC, length, and data type. After the checks and read operation are complete, it sets the corresponding status in the status field of the Flag in the TFLV structure. For status values, refer to “Appendix - Code Table - TFLV Result - Status Identifier” . The adjusted request-packet data body is returned as the response-packet data body.

The format of the Configuration Parameter Request Response Packet is as follows:



The TAG values that TFLV can carry are based on the “TAG Definition for Each Device Type” ..

11. Set Configuration Parameter Request Packet(0x06)

When the Server needs to modify configuration parameters on the Client, the Server should actively send a Set Configuration Parameter Request Packet to the Client.

Server:

The Set Configuration Parameter Request Packet requires the Client to wait for a response packet. Receipt of the response packet indicates successful transmission. If no response is received within the timeout period, resend the Set Configuration Parameter Request Packet. If the retransmission count is exceeded, transmission is considered unsuccessful; disconnect and reconnect.

The format of the Configuration Parameter Request Packet as follows:



The TAG values that TFLV can carry are based on the “TAG Definition for Each Device Type” ..

12. Set Configuration Parameter Request Response

Packet(0x86)

When the Client receives a Set Configuration Parameter Request Packet, it parses the packet and checks the CRC, length, and data type. After the checks and setting operation are complete, it sets the corresponding status in the status field of the Flag in the TFLV structure. For status values, refer to “Appendix - Code Table - TFLV Result - Status Identifier” . The adjusted request-packet data body is returned as the response-packet data body.

The format of the Configuration Parameter Request Response Packet is as follows:



The TAG values that TFLV can carry are based on the “TAG Definition for Each Device Type” ..

13. Status Query Request Packet(0x07)

When the Server needs to read status information from the Client, the Server should actively send a Status Query Request Packet to the Client.

In the TFLV structure, set Length and Type according to the corresponding values for the Tag, and fill Data Value with 0x00 for the corresponding length.

Server:

The status query request packet requires waiting for a response. Receipt of the response indicates successful transmission. If no response is received within the timeout period, resend the request packet. If the retransmission count is exceeded, transmission is considered unsuccessful and the connection is disconnected and re-established.

The format of the Status Query Request Packet is as follows:



The TAG values that TFLV can carry are based on the “TAG Definition for Each Device Type” ..

14. Status Query Request Response Packet(0x87)

When the Client receives a Status Query Request Packet, it parses the packet and checks the CRC, length, and data type. After the checks and read operation are complete, it sets the corresponding status in the status field of the Flag in the TFLV structure. For status values, refer to “Appendix - Code Table - TFLV Result - Status Identifier” . The adjusted request-packet data body is returned as the response-packet data body.

The format of the Status Query Request Response Packet is as follows:



The TAG values that TFLV can carry are based on the “TAG Definition for Each Device Type” ..

15. Device Control Request Packet(0x08)

When the Server needs to control the device, the Server should actively send a Device Control Request Packet to the Client.

In the TFLV structure, set Length and Type according to the corresponding values for the Tag, and fill Data Value with 0x00 for the corresponding length.

The format of the Device Control Request Packet is as follows:



The TAG values that TFLV can carry are based on the “TAG Definition for Each Device Type” ..

16. Device Control Request Response Packet(0x88)

When the Receiver receives a Device Control Request Packet, it parses the packet and checks the CRC, length, and data type. After the checks and read operation are complete, it sets the corresponding status in the status field of the Flag in the TFLV structure. For status values, refer to “Appendix - Code Table - TFLV Result - Status Identifier” . The adjusted data body of the original request packet is returned as the response-packet data body.

For the operations performed by each control-command request packet, refer to the command TAG description.

The format of the Device Control Request Response Packet is as follows:



The TAG values that TFLV can carry are based on the “TAG Definition for Each Device Type” ..

17. Read Configuration Parameter Information Request Packet(0x0A)

When the Server needs to read the Client’ s configuration parameter information, including which configuration parameters the device supports, the Server should actively send a Read Configuration Parameter Information Request Packet to the Client.

Server:

The Read Configuration Parameter Information Request Packet requires waiting for a response packet. Receipt of the response packet indicates successful transmission. If no response is received within the timeout period, resend the Read Configuration Parameter Information Request Packet. If the retransmission count is exceeded, transmission is considered unsuccessful; disconnect and reconnect.

The format of the Read Configuration Parameter Information Request Packet is as follows:

None

18. Read Configuration Parameter Information Request Response Packet(0x8A)

When the Client receives a Read Configuration Parameter Information Request Packet, it returns the list of configuration-parameter TAGs supported by the device.



Note: Because the number of Client parameters is not fixed, one request may require multiple response packets.

The returned data body is formatted as follows:



TAG values are defined in “TAG Definition for Each Device Type” .

19. Read Status Parameter Information Request Packet(0x0B)

When the Server needs to read the Client’ s status parameter information, including which status parameters the device supports, the Server should actively send a Read Status Parameter Information Request Packet to the Client.

Server:

The Read Status Parameter Information Request Packet requires waiting for a response packet. Receipt of the response packet indicates successful transmission. If no response is received within the timeout period, resend the Read Status Parameter Information Request Packet. If the retransmission count is exceeded, transmission is considered unsuccessful; disconnect and reconnect.

The format of the Read Status Parameter Information Request Packet is as follows:

None

20. Read Status Parameter Information Request Response Packet(0x8B)

When the Client receives a Read Status Parameter Information Request Packet, it returns the list of status-parameter TAGs supported by the device.



Note: Because the number of Client parameters is not fixed, one request may require multiple response packets.

The returned data body is formatted as follows:



TAG values are defined in “TAG Definition for Each Device Type” .

21. Read Control Parameter Information Request Packet(0x0C)

When the Server needs to read the Client’ s control-command information, including which Control Parameters the device supports, the Server should actively send a Read Control Parameter Information Request Packet to the Client.

Server:

The Read Control Parameter Information Request Packet requires waiting for a response packet. Receipt of the response packet indicates successful transmission. If no response is received within the timeout period, resend the Read Control Parameter Information Request Packet. If the retransmission count is exceeded, transmission is considered unsuccessful; disconnect and reconnect.

The format of the Read Control Parameter Information Request Packet is as follows:

None

22. Read Control Parameter Information Request Response Packet(0x8C)

When the Client receives a Read Control Parameter Information Request Packet, it returns the list of control-command TAGs supported by the device.



Note: Because the number of Client parameters is not fixed, one request may require multiple response packets.

The returned data body is formatted as follows:



TAG values are defined in “TAG Definition for Each Device Type” .

23. Read Event Parameter Information Request Packet(0x0D)

When the Server needs to read the Client' s event parameter information, including which event parameters the device supports, the Server should actively send a Read Event Parameter Information Request Packet to the Client.

Server:

The Read Event Parameter Information Request Packet requires waiting for a response packet. Receipt of the response packet indicates successful transmission. If no response is received within the timeout period, resend the Read Event Parameter Information Request Packet. If the retransmission count is exceeded, transmission is considered unsuccessful; disconnect and reconnect.

The format of the Read Event Parameter Information Request Packet is as follows:

None

24. Read Event Parameter Information Request Response Packet(0x8D)

When the Client receives a Read Event Parameter Information Request Packet, it returns the list of event-parameter TAGs supported by the device.



Note: Because the number of Client parameters is not fixed, one request may require multiple response packets.

The returned data body is formatted as follows:



TAG values are defined in “TAG Definition for Each Device Type” .

25. Upgrade Startup Packet(0x0E)

The data body format of the upgrade startup packet is as follows. File size is in bytes, the file

check code is the cumulative sum of the file data, and the timeout is in seconds:

File Size [4]	File Checksum [4]	Upgrade Timeout [2]	Reserved [10]
---------------	-------------------	---------------------	---------------

26. Upgrade Startup Response Packet(0x8E)

The data body format of the upgrade startup response packet is as follows:

None

27. Upgrade Data Transfer Packet(0x0F)

The data body format of the upgrade data transfer packet is as follows. After all packets have been transferred, send a packet whose upgrade-packet offset equals the firmware size and whose upgrade-packet data length is 0 as the end marker:

Packet Offset [4]	Upgrade Packet Data [n]
-------------------	-------------------------

28. Upgrade Data Transfer Response Packet(0x8F)

None

8. Appendix

8.1. Code Table

8.1.1. Status Identifier

Value	Description
0x00	Normal packet ID
0x01	Wrong protocol version number
0x03	Abnormal subpacket sequence number
0x04	Wrong device type
0x05	Abnormal device number
0x06	Wrong packet type
0x07	Abnormal length
0x08	Upgrade failed
0x7F	Other errors

8.1.2. Function Identifier

Value	Description
Bit0	Reserve
Bit1	0: Must respond; 1: No need to respond
Bit2	0: Application data is not encrypted; 1: Application data is encrypted (AES128-CFB)

8.1.3. Device Type

Value	Description
0x00 None	0x00 None
0x01 Smart Pill Box V1.2	0x01 Smart Pill Box V1.2
0x02 Smart Pill Box V2	0x02 Smart Pill Box V2

8.1.4. Device ID

Value (binary)	Description
bit63~bit62: 00: MAC 01: IMEI 10: Reserved 11: Reserved	Device identification type. The MAC is filled in byte order, and the high bits are filled with 0s. The IMEI number is filled in according to the 8421-BCD code.
bit61	Reserve. Filled with 0.
bit60	Reserve. Filled with 0.
Bit59~bit0	Device identification number

8.1.5. TFLV Structure - Status Identification

Value (binary)	Description
000	success
001	TAG is invalid
010	Invalid parameter type
011	The length of TFLV does not

	match the actual length of the value
100	Illegal value
101	operation failed
111	other

8.1.6. TFLV Structure - Parameter Type

Value (binary)	V String representation	Description
00000	UNKONW	unknown
00001	INT8S	Signed 8-bit integer
00010	INT8U	unsigned 8-bit integer
00011	INT16S	Signed 16-bit integer
00100	INT16U	Unsigned 16-bit integer
00101	INT32S	Signed 32-bit integer
00110	INT32U	Unsigned 32-bit integer
00111	INT64S	Signed 64-bit integer
01000	INT64U	Unsigned 64-bit integer
01001	FLOAT	Single precision floating point number
01010	DOUBLE	Double precision floating point number
01011	STRING	string
01100	ARRAY	byte array

8.2. CRC Algorithm C Code

```
typedef unsigned char INT8U;
typedef unsigned short int INT16U;
```

```

/*****
* Function name: CalcCRC
* Function description: single-byte CRC
* Tables accessed:
```

* Tables modified:

* Input parameters:

INT8U ucData : Character

INT16U usCRcIn : Previous CRC value

* Output parameters:

* Return value:

Returns the CRC value

* OtherDescription:

* Revision date, version, author, and revision description

* -----

*****/

```
static INT16U CalcCRC(INT8U ucData, INT16U usCRcIn)
{
    INT16U usIndex = 0;
    INT16U usCheck = 0;

    usCRcIn = usCRcIn ^ ucData;

    for (usIndex = 0; usIndex < 8; ++usIndex)
    {
        usCheck = usCRcIn & 1;
        usCRcIn = usCRcIn >> 1;
        usCRcIn = usCRcIn & 0x7FFF;

        if (1 == usCheck)
        {
            usCRcIn = usCRcIn ^ 0xA001;
        }
        usCRcIn = usCRcIn & 0xFFFF;
    }

    return usCRcIn;
}
```

/*****

* Function name: CRC16

* Function description: CRC16

* Tables accessed:

* Tables modified:

* Input parameters:

UCHAR *pucDst : Destination buffer address

INT16U usDstLen : Maximum destination buffer length

* Output parameters:

* Return value:

Returns the length of the processed data

* OtherDescription:

* Revision date, version, author, and revision description

* -----

*****/

```
INT16U CRC16(INT8U *pucSrc, INT16U usSrcLen)
```

```
{
```

```
    INT16U usCRC = 0;
```

```
    INT16U usIndex = 0;
```

```
    INT8U *pucTemp = NULL;
```

```
    usCRC = 0xFFFF;
```

```
    pucTemp = pucSrc;
```

```
    for (usIndex = 0; usIndex<usSrcLen; ++usIndex)
```

```
    {
```

```
        usCRC = CalcCRC(*pucTemp, usCRC);
```

```
        ++pucTemp;
```

```
    }
```

```
    return usCRC;
```

```
}
```

8.3. TAG Definition - Device Type 01

TAG	Description	Type	Length	Maximum	Minimum	Value Range
Configuration Parameters						
0x1001	Language	INT8U	-	1	0	0: Default 1: English
0x1002	Date format	INT8U	-	2	0	0: Year/Month/Day 1: Day/Month/Year 2: Month/Day/Year
0x1003	Time format	INT8U	-	1	0	0: 24-hour clock 1: 12-hour clock
0x1004	Alarm start year	INT16U	-	2100	2000	
0x1005	Alarm start month	INT8U	-	12	1	
0x1006	Alarm start day	INT8U	-	31	1	
0x1007	Alarm end year	INT16U	-	2100	2000	
0x1008	Alarm end month	INT8U	-	12	1	
0x1009	Alarm end day	INT8U	-	31	1	
0x1011	Alarm duration	INT16U	-	3600	0	(Reserved for future use) Unit: minutes 0 means continuous ringing
0x1012	Alarm ringtone type	INT8U	-	4	0	0: None 1: Ringtone 1 2: Ringtone 2 3: Ringtone 3 4: Ringtone 4
0x1013	Alarm ringtone volume	INT8U	-	2	0	0: High 1: Medium 2: Low 3: Mute
0x1014	Automatic Time Calibration	INT8U	-	1	0	0: Disabled 1: Enabled
0x1015	Time Zone	INT16S	-	+1400	-1200	
0x1016	Device ID	INT64U	8	-	-	Reserved for future use
0x1017	Medication	INT32U	-	86400	0	0 means that the

	Retrieval Timeout					device default value is used
0x1018	Missed Medication Retrieval Time	INT32U	-	86400	0	0 means that the device default value is used
0x1019	Early Medication Retrieval Validity Period	INT32U	-	3600	0	Reserved for future use 0 means that early medication retrieval is disabled
0x101A	Current Medication Compartment Number	INT8U	-	28	0	Reserved for future use
0x101B	Total Number of Medication Compartments	INT8U	-	29	29	Reserved for future use
0x101C	Loaded Medication Compartments	INT8U	-	28	0	
0x101D	Remaining Medication Compartments	INT8U	-	28	0	Reserved for future use
0x1021	Alarm1Hour	INT8U	-	23	0	
0x1022	Alarm2Hour	INT8U	-	23	0	
0x1023	Alarm3Hour	INT8U	-	23	0	
0x1024	Alarm4Hour	INT8U	-	23	0	
0x1025	Alarm5Hour	INT8U	-	23	0	
0x1026	Alarm6Hour	INT8U	-	23	0	
0x1027	Alarm7Hour	INT8U	-	23	0	
0x1028	Alarm8Hour	INT8U	-	23	0	
0x1029	Alarm9Hour	INT8U	-	23	0	
0x1031	Alarm1Minute	INT8U	-	59	0	
0x1032	Alarm2Minute	INT8U	-	59	0	
0x1033	Alarm3Minute	INT8U	-	59	0	
0x1034	Alarm4Minute	INT8U	-	59	0	
0x1035	Alarm5Minute	INT8U	-	59	0	
0x1036	Alarm6Minute	INT8U	-	59	0	
0x1037	Alarm7Minute	INT8U	-	59	0	
0x1038	Alarm8Minute	INT8U	-	59	0	
0x1039	Alarm9Minute	INT8U	-	59	0	
0x1051	Do Not Disturb	INT8U	-	1	0	0: Off

	Switch					1: On
0x1052	Do Not Disturb start hour	INT8U	-	23	0	
0x1053	Do Not Disturb start minute	INT8U	-	59	0	
0x1054	Do Not Disturb end hour	INT8U	-	23	0	
0x1055	Do Not Disturb end minute	INT8U	-	59	0	
0x1061	Emergency Call Trigger Time	INT16U	-	3600	0	Unit: seconds; an emergency call is triggered after inversion continues for N seconds; 0 disables this function.
0x1062	Key Tone Switch	INT8U	-	1	0	0: Off 1: On
0x1063	Alarm Ring Pause Duration	INT16U	-	1440	0	Unit: minutes; ringing resumes after N minutes of pause; 0 mutes the current alarm.

TAG	Description	Type	Length	Maximum	Minimum	Value Range
General Status Parameters						
0x8001	Device Type	INT8U	-	255	1	Deprecated 0: Unknown 1: Smart Pill Box V1.2
0x8002	Version	INT16U	-	65535	0	
0x8003	Packet Size	INT16U	-	1400	0	Unit: bytes
0x8004	Heartbeat packet interval	INT16U	-	65535	0	Unit: seconds
0x8005	Data Encryption	INT8U	-	1	0	0: No 1: Yes
0x8006	Response Timeout	INT16U	-	120	10	Unit: seconds
0x8007	Retransmission Count After Timeout	INT8U	-	10	0	

0x8008	Parameter Synchronization Identifier	INT8U	-	1	0	0: Non-forced synchronization 1: Forced synchronization
0x8009	SIM Card CCID	STRING	20	-	-	
0x800A	User Binding Relationship	INT8U	-	1	0	0: Not forced 1: Forced
0x800B	Online Hold Time	INT16U	-	65535	0	Unit: seconds; enters sleep mode after the online hold time; 0 means no sleep
0x8081	Additional Function Identifier	INT32U	-	0xFFFFFFFF	0	Bit0: Server switching Bit1: Bluetooth OTA
0x8082	Custom Number	INT32U	-	0xFFFFFFFF	0	Refer to the Custom Number description file for the value definition
Status Parameters						
0x8101	Medication Status	INT8U	-	2	0	0: Normal 1: Medication low 2: No medication
0x8103	Battery Level	INT8U	-	100	1	Battery percentage; 1 means 1%, 100 means 100%; not displayed while charging?
0x8104	Battery Status	INT8U	-	4	0	0: Normal 1: Fully charged 2: Low battery 3: Charging 4: No battery
0x8105	Pill Tray ID	STRING	20	-	-	Deprecated
0x8106	Low battery voltage	INT8U	-	1	0	Deprecated 0: Normal battery voltage 1: Low battery voltage
0x8107	Lid Status	INT8U	-	1	0	0: Lid closed 1: Lid open
0x8108	Pill Tray Rotation Fault	INT8U	-	1	0	0: Normal 1: Fault
0x8109	DC Power	INT8U	-	1	0	0: DC power

	Status					disconnected 1: DC power supplied
0x810A	Wi-Fi signal strength	INT16S	-	300	-300	
0x810B	GSM signal strength	INT16S	-	300	-300	
0x810C	Wi-Fi signal level	INT8U	-	4	0	
0x810D	GSM signal level	INT8U	-	4	0	
0x811A	Current Medication Compartment Number	INT8U	-	28	0	
0x811B	Total Number of Medication Compartments	INT8U	-	29	29	
0x811C	Loaded Medication Compartments	INT8U	-	28	0	Reserved for future use
0x811D	Remaining Medication Compartments	INT8U	-	28	0	
0x8122	Pill Tray Reset Fault	INT8U	-	1	0	0: Normal 1: Fault
0x8123	Emergency Call Status	INT8U	-	1	0	0: Normal 1: Emergency state
0x8131	Alarm 1 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 4: Timed out while waiting for dispensing 6: Missed medication 7: Medication taken
0x8132	Alarm 2 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing

						4: Timed out while waiting for dispensing 6: Missed medication 7: Medication taken
0x8133	Alarm 3 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 4: Timed out while waiting for dispensing 6: Missed medication 7: Medication taken
0x8134	Alarm 4 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 4: Timed out while waiting for dispensing 6: Missed medication 7: Medication taken
0x8135	Alarm 5 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 4: Timed out while waiting for dispensing 6: Missed medication 7: Medication taken
0x8136	Alarm 6 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 4: Timed out while waiting for

						dispensing 6: Missed medication 7: Medication taken
0x8137	Alarm 7 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 4: Timed out while waiting for dispensing 6: Missed medication 7: Medication taken
0x8138	Alarm 8 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 4: Timed out while waiting for dispensing 6: Missed medication 7: Medication taken
0x8139	Alarm 9 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 4: Timed out while waiting for dispensing 6: Missed medication 7: Medication taken
0x8140	Pill Tray Rotation Status	INT8U	-	1	0	0: None 1: Rotating
0x8141	Medication Pause Status	INT8U	-	1	0	0: Normal 1: Paused

TAG	Description	Type	Length	Maximum	Minimum	Value Range
General Control Parameters						
0xA001	Device Restart	INT8U	-	1	0	0: Immediate restart 1: Delayed restart
0xA002	Restore Factory Settings	INT8U	-	0	0	
0xA003	Cancel forced synchronization	INT8U	-	0	0	
0xA004	Device Re-registration	INT8U	-	0	0	
0xA011	Set Heartbeat Interval	INT16U	-	65535	0	Unit: seconds
0xA012	Set Online Hold Time	INT16U	-	65535	0	Reserved for future use Unit: seconds; enters sleep mode after the online hold time; 0 means no sleep
0xA021	Set Server IP	STRING	16	-	-	
0xA022	Set Server Domain Name	STRING	64	-	-	
0xA023	Set Server Port	INT16U	-	65535	0	
Control Parameters						
0xA101	Device Time Calibration	STRING	20	-	-	2001-01-02T20:05:04
0xA102	Mute Device	INT8U	-	1	1	Mute Current Alarm
0xA103	Pill Tray Reset	INT8U	-	1	1	
0xA121	Set Wi-Fi SSID	STRING	32	-	-	Reserved for future use
0xA122	Set Wi-Fi password	STRING	32	-	-	Reserved for future use
0xA123	Early Medication Retrieval	INT8U	-	0	0	
0xA124	Rotate to Specified Medication Compartment	INT8U	-	28	0	0 means no rotation; a value greater than 0 rotates to the corresponding medication compartment
0xA125	Medication Pause Duration	INT8U	-	255	0	0 means normal; a value greater than 0

	Setting					prohibits medication for the specified number of minutes
--	---------	--	--	--	--	--

TAG	Description	Type	Length	Maximum	Minimum	Value Range
General Event Parameters						
0xC001	Exit Request	INT8U	-	1	0	Reserved for future use 0: Device-initiated exit 1: Sleep-mode exit
Event Parameters						
0xC101	Time Calibration Request	INT16S	-	1400	-1200	
0xC201	Medication Retrieval Record	INT8U	-	5	0	0: Waiting for medication retrieval 1: On-time retrieval successful 2: Retrieval wait timed out 3: Retrieval after timeout successful 4: Medication not retrieved 5: Early retrieval successful
0xC202	Alarm Time	STRING	20	-	-	2001-01-02T20:05:04
0xC203	Pill Tray ID	STRING	20	-	-	Reserved for future use
0xC204	Current Medication Compartment Number	INT8U	-	28	0	
0xC205	Pill Tray Reset Report	INT8U	-	2	0	Deprecated 0: Pill tray reset started 1: Pill tray reset successful 2: Pill tray reset

						failed
0xC206	Device Startup Report	INT8U	-	0	0	Deprecated
0xC207	Medication Retrieval Trigger Identifier	INT8U	-	2	0	0: On-time medication retrieval 1: Early medication retrieval 2: Delayed medication retrieval
0xC208	Medication Retrieval Time	STRING	20	-	-	2001-01-02T20:05:04
0xC301	Pill Tray Reset Report	INT8U	-	2	0	Reserved for future use 0: Pill tray reset started 1: Pill tray reset successful 2: Pill tray reset failed
0xC401	Network Configuration Result Report	INT8U	-	1	0	Reserved for future use 0: Success 1: Failure
0xC402	Network Configuration Identifier	INT16U	-	65535	0	Reserved for future use
0xC403	Network Configuration Method	INT8U	-	1	0	Reserved for future use 0: Quick network configuration 1: Manual network configuration
0xC501	Factory Reset Report	INT8U	-	1	0	Reserved for future use 0: Local operation 1: Remote operation
0xC601	Device Startup Report	INT8U	-	0	0	Reserved for future use

8.4. TAG Definition - Device Type 02

TAG	Description	Type	Length	Maximum	Minimum	Value Range
Configuration Parameters						
0x1001	Language	INT8U	-	1	0	0: Default 1: English
0x1002	Date format	INT8U	-	2	0	0: Year/Month/Day 1: Day/Month/Year 2: Month/Day/Year
0x1003	Time format	INT8U	-	1	0	0: 24-hour clock 1: 12-hour clock
0x1004	Alarm start year	INT16U	-	2100	2000	
0x1005	Alarm start month	INT8U	-	12	1	
0x1006	Alarm start day	INT8U	-	31	1	
0x1007	Alarm end year	INT16U	-	2100	2000	
0x1008	Alarm end month	INT8U	-	12	1	
0x1009	Alarm end day	INT8U	-	31	1	
0x100A	Alarm Valid-Date Switch	INT8U	-	1	0	0: Off (the alarm is always valid) 1: On (the alarm is valid only within the valid dates)
0x100B	Key Tone Switch	INT8U	-	1	0	0: Off 1: On
0x100C	Safety Mode (Child Lock/Key Lock) Switch	INT8U	-	1	0	0: Off 1: On
0x100D	Early Medication Retrieval Switch	INT8U	-	1	0	0: Off 1: On
0x100E	Emergency Call Switch	INT8U	-	3	0	0: Off 1: On
0x1012	Alarm Ringtone Type	INT8U	-	3	0	0: None 1: Ringtone 1 2: Ringtone 2 3: Ringtone 3 4: Ringtone 4
0x1013	Volume Level	INT8U	-	3	0	0: High

						1: Medium 2: Low 3: Mute
0x1014	Automatic Time Calibration Switch	INT8U	-	1	0	0: Disabled 1: Enabled
0x1015	Time Zone	INT16S	-	+1400	-1200	
0x1017	Medication Retrieval Warning Time	INT32U	-	86400	0	0 means that the device default value is used
0x1018	Missed Medication Retrieval Time	INT32U	-	86400	0	0 means that the device default value is used
0x1019	Missed-medication dispensing switch	INT8U	-	1	0	Whether dispensing is allowed after medication retrieval is missed 0: Disabled (not allowed) 1: Enabled (allowed)
0x101A	Current Medication Compartment Number	INT8U	-	28	0	Reserved for future use
0x101C	Loaded Medication Compartments	INT8U	-	28	0	
0x101D	Low-Medication Warning Compartment Count	INT8U	-	28	0	0 means no warning
0x1021	Alarm1Hour	INT8U	-	23	0	
0x1022	Alarm2Hour	INT8U	-	23	0	
0x1023	Alarm3Hour	INT8U	-	23	0	
0x1024	Alarm4Hour	INT8U	-	23	0	
0x1025	Alarm5Hour	INT8U	-	23	0	
0x1026	Alarm6Hour	INT8U	-	23	0	
0x1027	Alarm7Hour	INT8U	-	23	0	
0x1028	Alarm8Hour	INT8U	-	23	0	
0x1029	Alarm9Hour	INT8U	-	23	0	
0x1031	Alarm1Minute	INT8U	-	59	0	
0x1032	Alarm2Minute	INT8U	-	59	0	
0x1033	Alarm3Minute	INT8U	-	59	0	

0x1034	Alarm4Minute	INT8U	-	59	0	
0x1035	Alarm5Minute	INT8U	-	59	0	
0x1036	Alarm6Minute	INT8U	-	59	0	
0x1037	Alarm7Minute	INT8U	-	59	0	
0x1038	Alarm8Minute	INT8U	-	59	0	
0x1039	Alarm9Minute	INT8U	-	59	0	
0x1041	Alarm1 Switch	INT8U	-	1	0	0: Off 1: On
0x1042	Alarm2 Switch	INT8U	-	1	0	0: Off 1: On
0x1043	Alarm3 Switch	INT8U	-	1	0	0: Off 1: On
0x1044	Alarm4 Switch	INT8U	-	1	0	0: Off 1: On
0x1045	Alarm5 Switch	INT8U	-	1	0	0: Off 1: On
0x1046	Alarm6 Switch	INT8U	-	1	0	0: Off 1: On
0x1047	Alarm7 Switch	INT8U	-	1	0	0: Off 1: On
0x1048	Alarm8 Switch	INT8U	-	1	0	0: Off 1: On
0x1049	Alarm9 Switch	INT8U	-	1	0	0: Off 1: On
0x1051	Do Not Disturb Switch	INT8U	-	1	0	0: Off 1: On
0x1052	Do Not Disturb start hour	INT8U	-	23	0	
0x1053	Do Not Disturb start minute	INT8U	-	59	0	
0x1054	Do Not Disturb end hour	INT8U	-	23	0	
0x1055	Do Not Disturb end minute	INT8U	-	59	0	
0x1063	Alarm Ring Pause Duration	INT16U	-	1440	0	Unit: minutes; ringing resumes after N minutes of pause; 0 mutes the current alarm.

TAG	Description	Type	Length	Maximum	Minimum	Value Range
General Status Parameters						
0x8002	Version	INT16U	-	65535	0	
0x8003	Packet Size	INT16U	-	1400	0	Unit: bytes
0x8004	Heartbeat packet interval	INT16U	-	65535	0	Unit: seconds
0x8005	Data Encryption	INT8U	-	1	0	0: No 1: Yes
0x8006	Response Timeout	INT16U	-	120	10	Unit: seconds
0x8007	Retransmission Count After Timeout	INT8U	-	10	0	
0x8008	Parameter Synchronization Identifier	INT8U	-	1	0	0: Non-forced synchronization 1: Forced synchronization
0x8009	SIM Card CCID	STRING	20	-	-	
0x800A	User Binding Relationship	INT8U	-	1	0	0: Not forced 1: Forced
0x800B	Online Hold Time	INT16U	-	65535	0	Unit: seconds; enters sleep mode after the online hold time; 0 means no sleep
0x8081	Additional Function Identifier	INT32U	-	0xFFFFFFFF	0	Bit0: Server switching Bit1: Bluetooth OTA Bit2: Remote upgrade
0x8082	Custom Number	INT32U	-	0xFFFFFFFF	0	Refer to the Custom Number description file for the value definition
Status Parameters						
0x8101	Medication Status	INT8U	-	2	0	0: Medication normal 1: Medication low 2: No medication
0x8102	Child Lock (Key Lock) Status	INT8U	-	1	0	0: Unlocked 1: Locked
0x8103	Battery Level	INT8U	-	100	1	Battery percentage;

						1 means 1%, 100 means 100%; not displayed while charging?
0x8104	Battery Status	INT8U	-	4	0	0: Normal 1: Fully charged 2: Low battery 3: Charging 4: No battery
0x8105	Do Not Disturb Status	INT8U	-	2	0	0: Off 1: On 2: Enabled and active
0x8106	Medication Cup Status	INT8U	-	1	0	0: Medication cup removed 1: Medication cup inserted
0x8107	Pill Tray Lock Status	INT8U	-	1	0	0: Unlocked 1: Locked
0x8109	DC Power Status	INT8U	-	1	0	0: Off 1: On
0x810A	Wi-Fi signal strength	INT16S	-	300	-300	
0x810B	GSM signal strength	INT16S	-	300	-300	
0x810C	Wi-Fi signal level	INT8U	-	3	0	
0x810D	GSM signal level	INT8U	-	3	0	
0x810E	Temperature	INT8S	-	-40	120	Unit: ° C
0x810F	Humidity	INT8U	-	100	0	Unit: %RH
0x8111	Temperature and Humidity Status	INT8U	-	1	0	0: Normal 1: Alarm
0x8112	Emergency Call Status	INT8U	-	1	0	0: Normal 1: Emergency call in progress
0x811A	Current Medication Compartment Number	INT8U	-	28	0	
0x811B	Total Number of Medication Compartments	INT8U	-	29	29	

0x811D	Remaining Medication Compartments	INT8U	-	28	0	
0x8121	Pill Tray Rotation Fault Status	INT8U	-	1	0	0: Normal 1: Fault
0x8122	Pill Tray Reset Fault Status	INT8U	-	1	0	0: Normal 1: Fault
0x8123	Medication Compartment Pusher Fault Status	INT8U	-	1	0	0: Normal 1: Fault
0x8124	Medication Compartment Door Fault Status	INT8U	-	1	0	0: Normal 1: Fault
0x8125	Key Fault Status	INT8U	-	1	0	0: Normal 1: Fault
0x8131	Alarm 1 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 3: Waiting for medication retrieval 4: Timed out while waiting for dispensing 5: Timed out while waiting for medication retrieval 6: Missed medication 7: Medication taken
0x8132	Alarm 2 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 3: Waiting for medication retrieval 4: Timed out while waiting for dispensing 5: Timed out while

						waiting for medication retrieval 6: Missed medication 7: Medication taken
0x8133	Alarm 3 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 3: Waiting for medication retrieval 4: Timed out while waiting for dispensing 5: Timed out while waiting for medication retrieval 6: Missed medication 7: Medication taken
0x8134	Alarm 4 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 3: Waiting for medication retrieval 4: Timed out while waiting for dispensing 5: Timed out while waiting for medication retrieval 6: Missed medication 7: Medication taken
0x8135	Alarm 5 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 3: Waiting for medication retrieval 4: Timed out while waiting for

						dispensing 5: Timed out while waiting for medication retrieval 6: Missed medication 7: Medication taken
0x8136	Alarm 6 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 3: Waiting for medication retrieval 4: Timed out while waiting for dispensing 5: Timed out while waiting for medication retrieval 6: Missed medication 7: Medication taken
0x8137	Alarm 7 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 3: Waiting for medication retrieval 4: Timed out while waiting for dispensing 5: Timed out while waiting for medication retrieval 6: Missed medication 7: Medication taken
0x8138	Alarm 8 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 3: Waiting for medication retrieval

						4: Timed out while waiting for dispensing 5: Timed out while waiting for medication retrieval 6: Missed medication 7: Medication taken
0x8139	Alarm 9 status	INT8U	-	7	0	0: None 1: Preparing to dispense medication 2: Waiting for dispensing 3: Waiting for medication retrieval 4: Timed out while waiting for dispensing 5: Timed out while waiting for medication retrieval 6: Missed medication 7: Medication taken

TAG	Description	Type	Length	Maximum	Minimum	Value Range
General Control Parameters						
0xA001	Device Restart	INT8U	-	1	0	0: Immediate restart 1: Delayed restart
0xA002	Restore Factory Settings	INT8U	-	0	0	
0xA003	Cancel forced synchronization	INT8U	-	0	0	
0xA004	Device Re-registration	INT8U	-	0	0	
0xA011	Set Heartbeat Interval	INT16U	-	65535	0	Unit: seconds
0xA012	Set Online Hold Time	INT16U	-	65535	0	Reserved for future use Unit: seconds; enters

						sleep mode after the online hold time; 0 means no sleep
0xA021	Set Server IP	STRING	16	-	-	
0xA022	Set Server Domain Name	STRING	64	-	-	
0xA023	Set Server Port	INT16U	-	65535	0	
Control Parameters						
0xA101	Device Time Calibration	STRING	20	-	-	2001-01-02T20:05:04
0xA102	Mute Device	INT8U	-	1	1	Mute Current Alarm
0xA103	Pill Tray Reset	INT8U	-	1	1	
0xA123	Early Medication Retrieval	INT8U	-	0	0	

TAG	Description	Type	Length	Maximum	Minimum	Value Range
General Event Parameters						
0xC001	Exit Request	INT8U	-	1	0	Reserved for future use 0: Device-initiated exit 1: Sleep-mode exit
Event Parameters						
0xC201	Medication Alarm ID	INT8U	-	8	0	0: Alarm1 1: Alarm2 2: Alarm3 3: Alarm4 4: Alarm5 5: Alarm6 6: Alarm7 7: Alarm8 8: Alarm9
0xC202	Medication Alarm Time	STRING	20	-	-	2001-01-02T20:05:04
0xC203	Current Medication Time	STRING	20	-	-	2001-01-02T20:05:04
0xC204	Medication	INT8U	-	28	0	

	Compartment Number					
0xC205	Medication Retrieval Method	INT8U	-	2	0	0: On-time medication retrieval 1: Early medication retrieval 2: Delayed medication retrieval
0xC206	Medication Result	INT8U	-	3	0	0: Medication taken on time 1: Medication taken late (after warning) 2: Abnormal medication (taken after a missed-dose event) 3: Medication missed